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REDLINE / RL-MB-01

CONCEPT STUDY - PROVISIONAL DIMENSIONS - DO NOT BUILD WITHOUT REVIEW

NET LIST: 33 NETS, ALL TERMINALS UNIQUE (CHECKED)

NET	CLASS	TERMINALS
VBUS	power	J1.VBUS U4.VBUS
VBAT	power	BT1.+ J3.1 U4.VBAT
VSYS	power	U4.VSYS U1.VDD U7.VIN D1.A
VCAP	power	U7.CAP C1.+
V1V8	power	U4.BUCK1 U1.VDD_GPIO U2.VDD U3.VDD U5.VCC U6.VCC J4.VREF
GND	ground	U4.GND U1.GND U2.GND U3.GND U5.GND U6.GND U7.GND C1.- +9
CC1	signal	J1.CC1 R1.1
CC2	signal	J1.CC2 R2.1
SPI_SCK	bus	U1.SPI_SCK U2.SCLK U5.SCLK
SPI_MOSI	bus	U1.SPI_MOSI U2.MOSI U5.SI
SPI_MISO	bus	U1.SPI_MISO U2.MISO U5.SO
CS_IMPACT	bus	U1.CS0 U2.CS
CS_BLACKBOX	bus	U1.CS1 U5.CS
I2C_SDA	bus	U1.TWI_SDA U3.SDA U4.SDA
I2C_SCL	bus	U1.TWI_SCL U3.SCL U4.SCL
INT_IMPACT	signal	U2.INT1 U1.GPIO_INT1
INT_MOTION	signal	U3.INT1 U1.GPIO_INT2
PMIC_INT	signal	U4.GPIO0 U1.GPIO_INT3
PWR_FAIL	signal	U7.PG U1.GPIO_PWRFAIL
RF_CELL	RF	U1.ANT ANT1.FEED
GNSS_ANT	RF	ANT2.FEED U6.IN
GNSS_RF	RF	U6.OUT U1.GPS
LNA_EN	signal	U1.COEX0 U6.EN
SIM_VCC	bus	U1.SIM_VCC J2.VCC
SIM_CLK	bus	U1.SIM_CLK J2.CLK
SIM_IO	bus	U1.SIM_IO J2.IO
SIM_RST	bus	U1.SIM_RST J2.RST
BUTTON	signal	SW1.1 U1.GPIO_BTN
TAMPER	signal	TS1.1 U1.GPIO_TAMPER
LED_SINK	signal	D1.K U4.LED0
SWDIO	bus	U1.SWDIO J4.SWDIO
SWDCLK	bus	U1.SWDCLK J4.SWDCLK
NRESET	signal	U1.RESET J4.RESET

CONNECTOR PIN-OUTS

J1 USB-C	PIN	NET
	CC1	CC1
	CC2	CC2
	GND	GND
	VBUS	VBUS

J2 eSIM MFF2	PIN	NET
	CLK	SIM_CLK
	GND	GND
	IO	SIM_IO
	RST	SIM_RST
	VCC	SIM_VCC

J3 Battery JST-PH 2	PIN	NET
	1	VBAT
	2	GND

J4 SWD pads	PIN	NET
	GND	GND
	RESET	NRESET
	SWDCLK	SWDCLK
	SWDIO	SWDIO
	VREF	V1V8

SW1 Button	PIN	NET
	1	BUTTON
	2	GND

TS1 Tamper switch	PIN	NET
	1	TAMPER
	2	GND

CABLE W1 (ONLY CABLE INSIDE)

BT1 Li-Po

2000 mAh

red: VBAT

black: GND

J3

JST-PH 2 pins, 50 mm. Check polarity before plugging.

POWER-UP SEQUENCE

1. VSYS from U4 (battery or USB-C).

2. VDD of U1 and ENABLE high.

3. BUCK1 starts 1.8 V more than 6 ms later.

4. VDD_GPIO then follows, sensors wake.

5. C1 charges through U7; PWR_FAIL goes high when full.

6. On a supply drop U7 keeps VSYS up and raises PWR_FAIL: last gasp.

PASSIVE PARTS

Decoupling, RF matching, pull-ups and level parts are added at routing; only R1 and R2 (USB-C CC pull-downs) are modelled here because they define the connector.

GENERAL NOTES

1. Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).

2. Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.

3. Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.

4. Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

WIRING AND NET LIST

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO.	SCALE	REV	SHEET
RL-EL-05	tables	0	5 / 5
DRAWN BY: M. ABAAQIL	DATE: 2026-10-05	MATERIAL: to be defined	

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